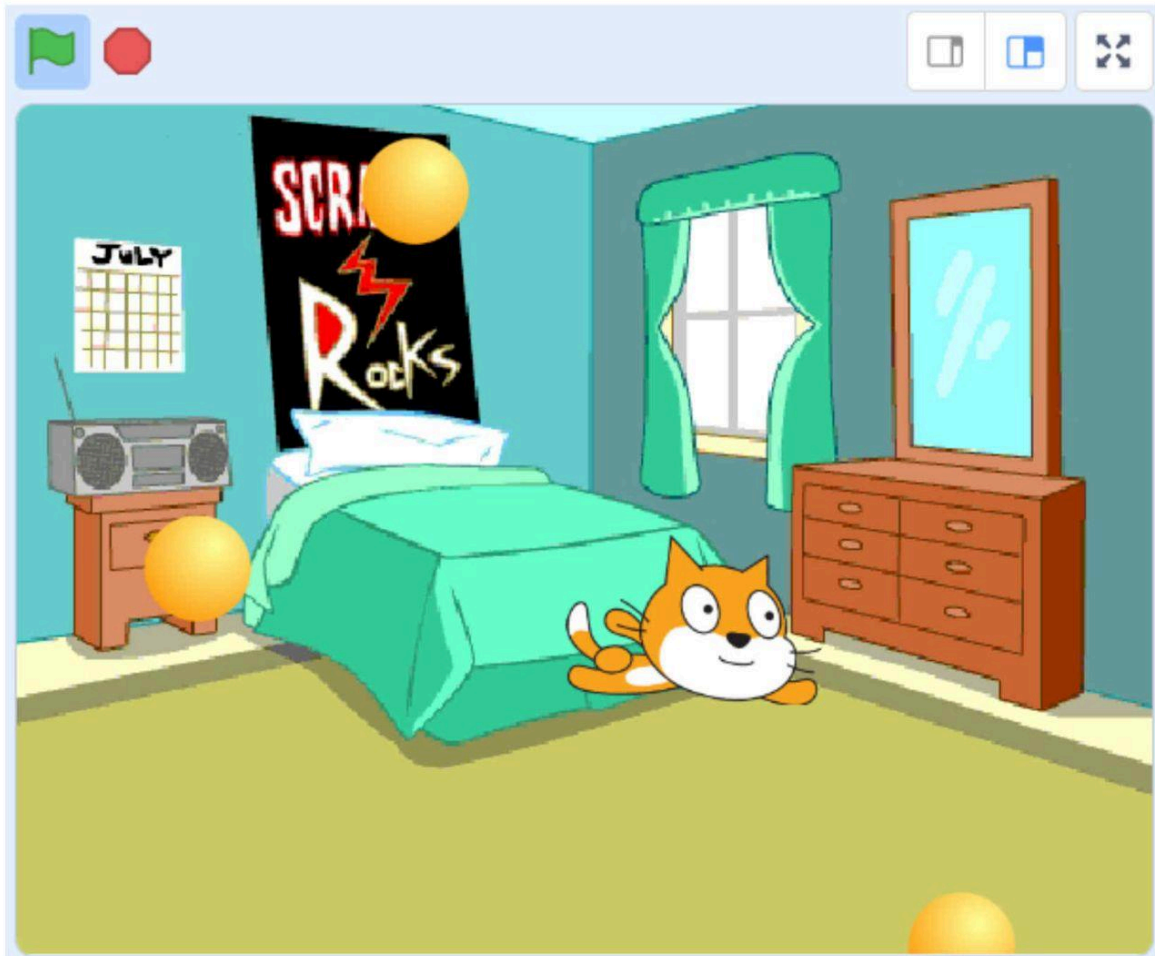


Unit 2

Motion Workshop



Catch the Ball

Unit 2: Motion Workshop

Save the file with a new name. Your new name should identify the project.

* Motion

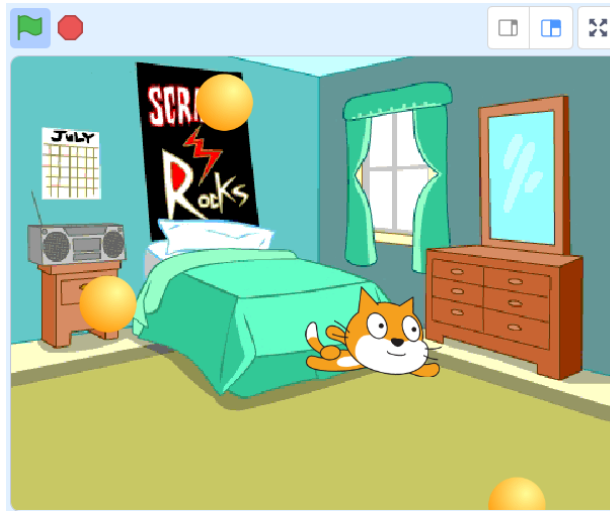
* Catch the Ball

* Project 2

Workshop summary

The Motion menu gives you control over your sprites. The blocks in this menu set the sprite's location and direction. In addition, there are blocks to move and glide.

In this workshop, students build a classic falling sprites game. The goal is to catch as many balls as possible with the flying cat.



New Terms

Interface Terms

coordinates, backdrop, x & y

Code Block Terms

move, glide, repeat until, hide, show, random, clone

Class Studio: <https://tinyurl.com/StchStudio01>

Video of Full Project: <https://vimeo.com/687493286>

This Worksheet: <https://tinyurl.com/MotionWSPDF>

Starter Project:

<https://tinyurl.com/StchStarter02>

Full Project:

<https://scratch.mit.edu/projects/656608586>

Mini-Challenges in this Lesson

Click on the mini-challenge link to go directly to that section.

[Mini-Challenge 1: Move the Cat with code](#)

[Mini-Challenge 2: Chase the computer mouse](#)

[Mini-Challenge 3: Add a new sprite](#)

[Mini-Challenge 4: Create a falling sprite](#)

[Mini-Challenge 5: Randomize the starting position a sprite](#)

[Mini-Challenge 6: Create a storm of falling sprites](#)

[Mini-Challenge 7: Hide the original sprite](#)

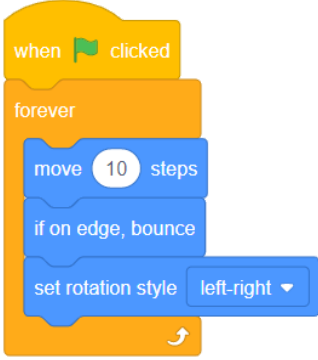
[Mini-Challenge 8: Make a falling sprite game](#)

Mini-Challenge 1: Move the Cat With Code

When the flag is clicked, the SCRATCH CAT flies around the bedroom chasing a BALL and bounces off the walls.

Procedure

1. Add a *when flag is clicked* block to signal the game to begin.
2. Add a forever loop.
3. Inside the forever loop.
 - a. Add a move (___) steps block and enter a small number (1-10) and run the program.
 - b. Add an “if on edge, bounce” block.
4. Control the sprite’s appearance with a “set rotation style [___]” block.

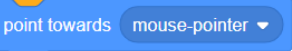
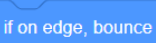
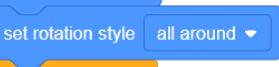
Menu	Code	Code block	Notes
Events	when flag clicked		Put code stack on the SCRATCH CAT .
Control	forever loop		
Motion	move (___) steps		
Motion	if on edge, bounce		
Motion	set rotation style [___]		

Mini-Challenge 2: Chase the Computer Mouse

Point the cat towards the mouse and move toward it.

Procedure

1. Inside the forever loop, add a Point towards [___] block.
2. Put this block in front of the other blocks in the forever loop.

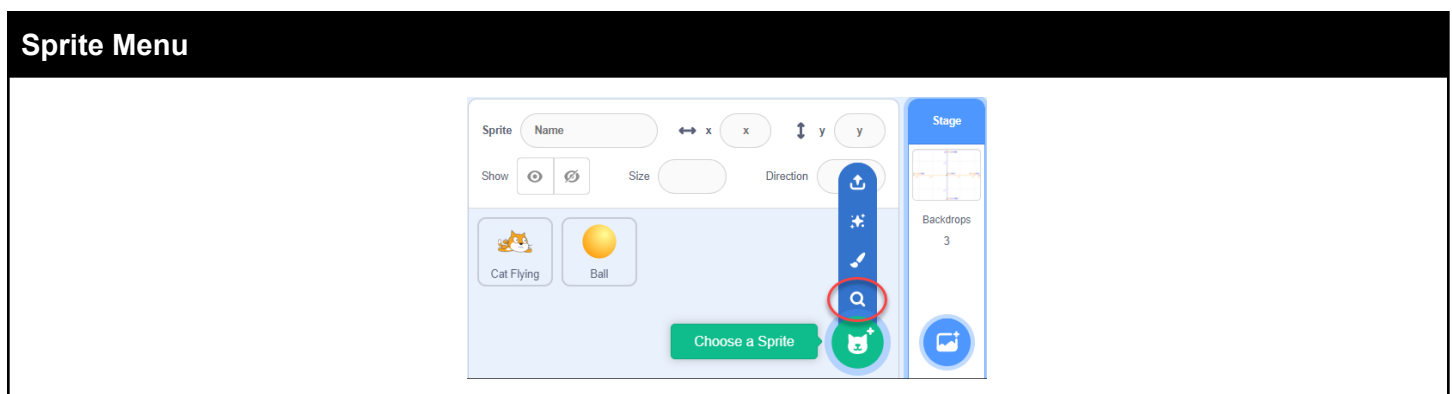
Menu	Code	Code block	Notes
Events	when clicked		Put code stack on the SCRATCH CAT .
Control	forever loop		
Motion	point towards [mouse-pointer]		
Motion	move () steps		
Motion	if on edge, bounce		
Motion	set rotation style []		

Mini-Challenge 3: Add a New Sprite

Add a new sprite that the CAT sprite will chase around the room.

Procedure

1. Add a new sprite that is small and compact (ball, star, clownfish, etc.)



Procedure

2. Drag the small sprite to three different locations on the stage. Each location should be in a different quarter of the stage.
3. For each location, observe the x and y settings in the sprite control panel.
4. Write down the x and y coordinates in each location on your note paper.
 - a. Write them like this: x: -100, y: 75
 - b. Your coordinates will probably be different than those in the picture.

Mini-Challenge 4: Create a Falling Sprite

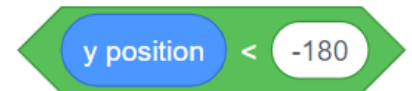
Make a sprite fall from the top of the stage over and over again.

Procedure

1. Switch to the little sprite used earlier to map the locations.
2. Drag your sprite to the top of the stage, as high as it will go. Your y position will probably be over 180.
3. Add a *when clicked* block and a forever loop.
4. Inside the forever loop, add a *go to x:() y:()* block. It should display the sprite's coordinates from step 3.
5. Add a *repeat until < >* block under the *go to x:() y:()* block.

The *repeat until* loop will be inside the *forever* loop!

- a. In the repeat diamond, add a *() < ()* operator block. Put a y position block in the first space and -180 in the second space
 - b. Add a *change y by ()* block in the repeat until loop. Set this block to a small negative number.
6. Run the program.



Menu	Code	Code block	Notes
Events	when clicked		Put code stack on the BALL sprite.
Control	forever loop		
Motion	go to x:() y:()		
Control	repeat until < >		Pro-tip: to make a sprite fall, you must change y by a negative number.
Operators	< (y position) < (-180) >		Note: We use -180 inside the less than operator.
Motion	change y by ()		This makes the repeat loop run until the sprite's y-coordinate is close to the bottom of the stage.

Mini-Challenge 5: Randomize the starting position of a sprite

Use random positions for the starting location of the falling sprite.

Procedure

- 1. Replace the *go to x:() y:()* block with a *go to random position* block.
- 2. Add a *set y to ()*; use a positive number close to the top of the stage.

Pro-tip: Go to random position sets the sprite's *x* and *y* to random numbers.
By adding the *set y to ()* block, the sprite will be at a random *x* location, but the same *y* location each time.

Menu	Code	Code block	Notes
Events	when clicked		Put code stack on the BALL sprite.
Control	forever loop		
Motion	go to (random position)		
Motion	set y to (174)		
Control	repeat until <__>		
Operator	< (y position) < (-180) >		
Motion	change y by ()		

Mini-Challenge 6: Create a Storm of Falling Sprites

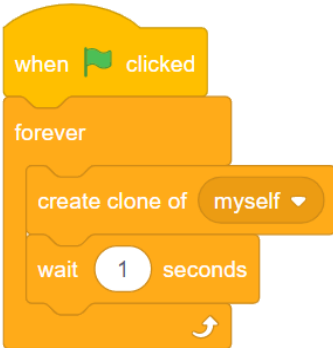
Use cloning to create a sprite storm.

Procedure

- 1. Drag all of the code out of the *forever loop* and place it on the code canvas.
- 2. In the *forever loop*, add a *create clone of myself* block.
- 3. Add a *wait 1-second* block after the clone block.



Pro-tip: You can control how many clones appear by changing the wait time.
If the wait amount is missing or too small, new clones will be created so fast that the stage will be filled with them!

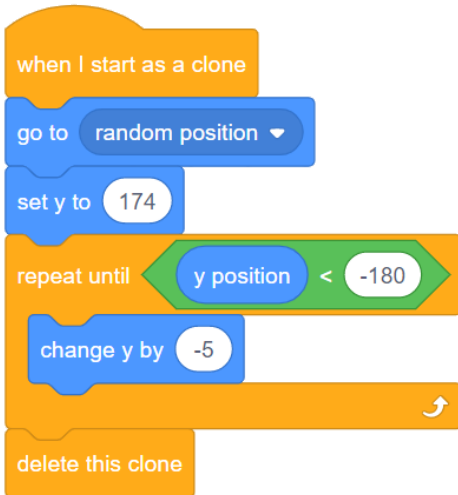
Menu	Code	Code block	Notes
Events	when clicked		Put code stack on the BALL sprite.
Control	forever loop		
Control	create clone of myself		
Control	wait (__) seconds		

Procedure

4. Add a *when I start as a clone* block to the code canvas.

Note: Since this is a hat block, you can't put this in the *when clicked* code stack. It is a new code stack.

5. Connect the blocks from #1 to the *when I start as a clone* block.
6. Add a *delete this clone* to the end of this code stack.
7. Run the program.

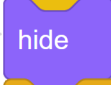
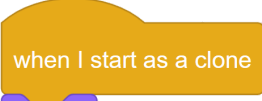
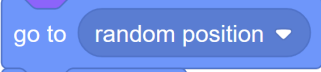
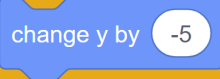
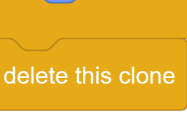
Menu	Code	Code block	Notes
Events	when I start as a clone		Put code stack on the BALL sprite.
Motion	go to (random position)		
Motion	set y to (174)		
Control	repeat until < __ >		
Operators	< (y position) < (-180) >		
Motion	change y by (__)		
Control	delete this clone		Pro-tip: We add the <i>delete this clone</i> block so that the clones don't pile up at the bottom of the stage.

Mini-Challenge 7: Hide the Original Sprite

Did you notice that your original BALL sprite is no longer moving? It is just sitting on the stage. That is because all of the movement blocks have been switched to the clone! We can't get rid of the sprite, but we can hide it.

Procedure

1. In the *when clicked* code stack, set the visibility to *hide*.
2. In the *when I start as a clone* code stack, set the visibility to *show*.

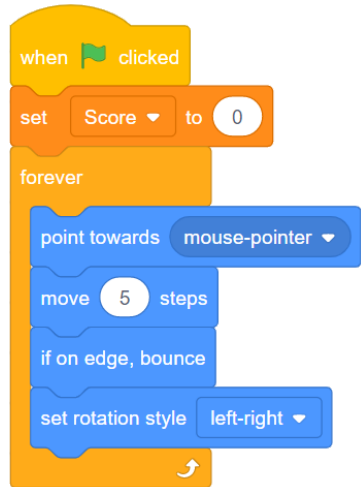
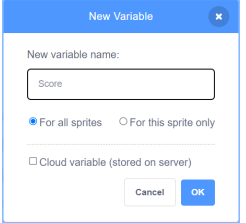
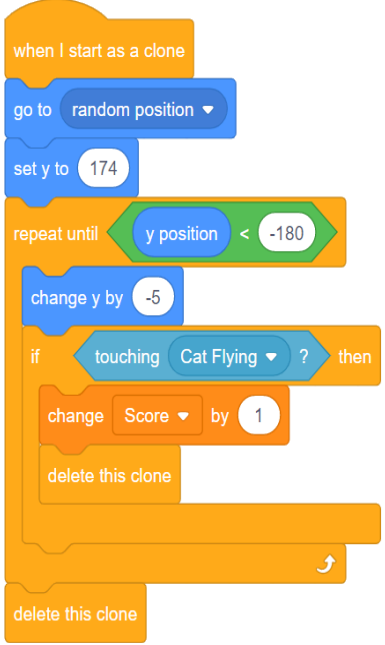
Menu	Code	Code block	Notes
Events	when clicked		Put code stack on the BALL sprite.
Looks	hide		
Control	forever loop		
Control	create clone of myself		
Control	wait (__) seconds		
Events	when I start as a clone		Put code stack on the BALL sprite.
Looks	show		
Motion	go to (random position)		
Motion	set y to (183)		
Control Motion/Operators	repeat until < __ > < (y position) < (-180) >		
Motion	change y by (__)		
Control	delete this clone		

Mini-Challenge 8: Make a falling sprite game

Let's add code to the CAT sprite so it can chase the falling sprites. Use a variable to keep score!

Procedure

1. Switch to the CAT sprite.
2. The original code from Mini-Challenge 3 should be present, so you can control the CAT.
3. Add a variable for **Score** using the Variables menu.
4. Add a set (___) above the *forever loop*.
5. Switch to the small sprite.
6. Inside the *repeat until* loop, add an *if, then* block.

Menu	Code	Code block	Notes
Events	when clicked		Put code stack on the SCRATCH CAT .
Variables	set (Score) to 0		
Control	forever loop		
Motion	point towards (mouse pointer)		
Motion	move (___) steps		
Motion	if on edge, bounce		
Motion	set rotation style [___]		
Events	when I start as a clone		Put code stack on the BALL sprite.
Motion	go to (random position)		
Motion	set y to (174)		
Control	repeat until < ___ >		
Operators	< (y position) < (-180) >		
Motion	change y by (___)		
Control	if touching (cat flying) then		
Variables	change Score by 1		
Control	delete this clone		
Control	delete this clone		